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Numerical Methods for High-Speed Flows

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Keywords

shock waves, shock-capturing schemes, energy conservation, numerical dissipation

Abstract

We review numerical methods for direct numerical simulation (DNS) and large-eddy simulation (LES) of turbulent compressible flow in the presence of shock waves. Ideal numerical methods should be accurate and free from numerical dissipation in smooth parts of the flow, and at the same time they must robustly capture shock waves without significant Gibbs ringing, which may lead to nonlinear instability. Adapting to these conflicting goals leads to the design of strongly nonlinear numerical schemes that depend on the geometrical properties of the solution. For low-dissipation methods for smooth flows, numerical stability can be based on physical conservation principles for kinetic energy and/or entropy. Shock-capturing requires the addition of artificial dissipation, in more or less explicit form, as a surrogate for physical viscosity, to obtain nonoscillatory transitions. Methods suitable for both smooth and shocked flows are discussed, and the potential for hybridization is highlighted. Examples of the application of advanced algorithms to DNS/LES of turbulent, compressible flows are presented.

FD: finite difference

DNS: direct numerical simulation

LES: large-eddy simulation

1. INTRODUCTION

Numerical simulation of high-speed flows has a long history, dating back to the beginning of the computer era (Courant et al. 1952, Godunov 1959, von Neumann & Richtmyer 1950). Several textbooks on numerical methods have appeared over the years (Hirsch 1988, Laney 1998, LeVecque 1990, Richtmyer & Morton 1967, Toro 2009). Important advancements have been made, but as illustrated by a recent comparative study (Johnsen et al. 2010), computational gasdynamics has not yet converged to an optimal computational strategy. The purpose of this review is to check the status of the discipline and systematize the enormous amount of material produced over the years, part of which was reviewed by Ekaterinaris (2005). Given the vastness of the subject, we mostly limit ourselves to analyzing the family of finite-difference (FD) schemes that are frequently used, especially in the academic community, for direct numerical simulation (DNS) and large-eddy simulation (LES) of compressible turbulent flows. Resolving the range of scales present in these flows requires numerical schemes that are accurate, robust, and efficient in terms of CPU requirements.

The reference physical model consists of the compressible Navier-Stokes equations for a calorically perfect gas, here written in integral form in Cartesian coordinates:

$$\frac{d}{dt} \int_V \mathbf{u} dV + \sum_{i=1}^3 \int_{\partial V} (\mathbf{f}_i - \mathbf{f}_i^v) n_i dS = 0, \quad (1)$$

where

$$\mathbf{u} = \begin{bmatrix} \rho \\ \rho u_j \\ \rho E \end{bmatrix}, \quad \mathbf{f}_i = \begin{bmatrix} \rho u_i \\ \rho u_i u_j + p \delta_{ij} \\ \rho u_i H \end{bmatrix}, \quad \mathbf{f}_i^v = \begin{bmatrix} 0 \\ \sigma_{ij} \\ \sigma_{ik} u_k - q_i \end{bmatrix}, \quad j = 1, 2, 3, \quad (2)$$

are the vector of conservative variables and the vectors of the convective and viscous fluxes in the i -th direction, respectively. Here ρ is the density, u_i is the velocity component in the i -th coordinate direction, p is the thermodynamic pressure, $E = e + \rho u^2/2$ is the total energy per unit mass, $e = RT/(\gamma - 1)$ is the internal energy per unit mass, $H = E + p/\rho$ is the total enthalpy, R is the gas constant, $\gamma = c_p/c_v$ is the specific heat ratio, σ_{ij} is the viscous stress tensor, and q_i is the heat flux vector. At high Mach numbers, the occurrence of very strong shock waves and of severe viscous heating may bring to light additional effects that are not incorporated in Equation 1, such as chemical reactions and nonideal thermodynamic behavior (real-gas effects). These are not addressed in the present review, and the interested reader is referred to specific publications (e.g., Bertin & Cummings 2006).

Under the assumption of smooth flow, the Navier-Stokes equations can be equivalently cast in differential form:

$$\frac{\partial \mathbf{u}}{\partial t} + \sum_{i=1}^3 \frac{\partial \mathbf{f}_i}{\partial x_i} = \sum_{i=1}^3 \frac{\partial \mathbf{f}_i^v}{\partial x_i}. \quad (3)$$

Neglect of molecular diffusion effects (i.e., setting $\mathbf{f}_i^v = 0$) leads to the Euler equations, which only incorporate the effects of macroscopic convection and molecular collisional effects through pressure forces. The Euler equations have several important mathematical properties that are illustrated in the books of Lax (1973), Majda (1984), Hirsch (1988), and LeVecque (1990). Some useful properties for the development of numerical methods are briefly recalled here.

First, the system of Euler equations can be cast in characteristic form, meaning that projection of the equations in any spatial direction gives rise to a system of coupled wave-like equations (Majda 1984). This motivates the study of the model one-dimensional (1D) scalar conservation law:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = \frac{\partial u}{\partial t} + a(u) \frac{\partial u}{\partial x} = 0, \quad (4)$$

which is used as a prototype for the development of numerical methods for hyperbolic equations. It can also be shown that the convective flux vectors are homogeneous functions of order one with respect to the vector of conservative variables, i.e., $\mathbf{f}_i = d\mathbf{f}_i/d\mathbf{u} \cdot \mathbf{u}$, a property that is useful for the design of upwind methods based on the flux vector splitting technique (Steger & Warming 1981).

Second, the Euler equations have the obvious property (as is clear from their integral form) that the integrals of ρ , ρu_i , and ρE over an arbitrary control volume can only vary because of flux through the boundaries. Under the assumption of smooth flow, combining the continuity and the momentum equations and integrating yield a balance equation for the kinetic energy $\rho u_k u_k/2$:

$$\frac{d}{dt} \int_V \rho u_k u_k/2 dV = - \int_{\partial V} (\rho u_k u_k/2 + p) u_i n_i dS + \int_V p \frac{\partial u_i}{\partial x_i} dV. \quad (5)$$

Equation 5 shows that the total kinetic energy only varies because of momentum flux through the boundary or to volumetric work of pressure forces (which is zero for incompressible flow), whereas the convective terms do not cause any net variation. This property has inspired numerical schemes based on the attempt to enforce kinetic energy preservation in the discrete sense. Additional conservation laws can be derived from the Euler equations for smooth flows, namely (Harten 1983b)

$$\frac{d}{dt} \int_V \rho g(s) dV + \int_{\partial V} \rho g(s) u_i n_i dS = 0, \quad (6)$$

where $g(s)$ is an arbitrary (but differentiable) function of the thermodynamic entropy $s = \log(p\rho^{-\gamma})$, which suggests the possibility to design numerical schemes that discretely preserve the integral of $\rho g(s)$. In the presence of shock waves, mechanical energy is transformed into heat (with subsequent production of entropy) even in the limit of vanishing viscosity, and energy/entropy conservation no longer holds.

Finally, Harten (1983b) showed that the change of variables $\mathbf{v} = \partial \rho g(s)/\partial \mathbf{u}$ (where the $\mathbf{v}$ are referred to as entropy variables) allows the Euler equations to be recast in the symmetrized form

$$\mathbf{P} \frac{\partial \mathbf{v}}{\partial t} + \sum_{i=1}^3 \mathbf{B}_i \frac{\partial \mathbf{v}}{\partial x_i} = 0, \quad (7)$$

where $\mathbf{P} = \partial \mathbf{u}/\partial \mathbf{v}$ and $\mathbf{B}_i = \partial \mathbf{f}_i/\partial \mathbf{v}$ are symmetric matrices, and $\mathbf{P}$ is positive definite. The symmetrized form of the equations has the important property that continuous energy estimates can be established (Olsson 1993, Olsson & Oliger 1994). Indeed, referring to a 1D bounded interval $[a, b]$, from Equation 7 it follows that

$$\frac{d}{dt} \int_a^b \mathbf{v} \cdot \mathbf{P} \mathbf{v} dx = -[\mathbf{v} \cdot \mathbf{B} \mathbf{v}]_a^b, \quad (8)$$

showing that the generalized energy on the left-hand side can only vary in time because of boundary contributions. Tadmor (1984) showed that validity of the energy estimate given in Equation 8 implies satisfaction of entropy conservation (Equation 6). The choice $g(s) = \beta \exp(s/(\beta(\gamma - 1)))$ (Harten 1983b) leads to simple transformations between conservative and entropy variables (Sandham et al. 2002), and $\mathbf{u}$, $\mathbf{f}_i$ happen to be homogeneous functions of degree β with respect to $\mathbf{v}$,

$$\mathbf{u}(\theta \mathbf{v}) = \theta^\beta \mathbf{u}(\mathbf{v}), \quad \mathbf{f}_i(\theta \mathbf{v}) = \theta^\beta \mathbf{f}_i(\mathbf{v}). \quad (9)$$

The solutions of Equation 1 exhibit a tendency to form steep gradients (shock waves and contact discontinuities), with thickness of the order of magnitude of the gas mean-free path, making shock resolution with numerical methods infeasible in most cases of practical interest. In the inviscid limit shock waves reduce to zero-measure transition zones, across which the Rankine-Hugoniot

DRP: dispersion relation preserving

jump conditions must be satisfied. In order for the class of weak solutions of the Euler equations to coincide with that of the Navier-Stokes equations in the inviscid limit, additional conditions (in the form of entropy inequalities) must be satisfied to ensure that the macroscopic effects of the small-scale diffusion processes are properly represented. For a thorough mathematical discussion of these issues, the reader may consult Lax (1973), Majda (1984), and LeVeque (1990).

To summarize, high-speed flows typically feature regions where the flow is smooth, and the governing equations in their differential form hold, interspersed by extremely thin regions, where the flow properties vary abruptly. A possible exception is the case of flows in which shocklets embedded in turbulent flow occur, associated with velocity fluctuations of the order of the sound speed (Lee et al. 1991). Apparently, even when this happens, their frequency and strength are not such to severely threaten the robustness and accuracy of numerical algorithms. Furthermore, the shocklets' thickness is found to scale on the Kolmogorov length, rather than the mean-free path (Samtaney et al. 2001), making their resolution possible in DNS. Therefore, it is not surprising that numerical methods for high-speed flows have specialized into two classes, one capable of dealing with smooth flows and the other with shock waves, each with quite different properties. Indeed, it is known that standard discretizations used for smooth flows cause (potentially dangerous) Gibbs oscillations in the presence of shock jumps, whereas typical methods used to regularize shock calculations exhibit excessive numerical viscosity. Schemes of the two families are illustrated in Sections 2 and 3, respectively. Tailored numerical methods for high-speed flows should be able to put together the favorable properties of the two basic families, which is frequently realized through their hybridization or blending, as also explained in Section 3. The following discussion mostly deals with the discretization of the convective derivatives that appear in Equation 3, as this is the main source of nonlinearity in the governing equations. As customary in the computational gasdynamics community, a method-of-lines approach is assumed, whereby time integration is handled separately, using ODE dedicated methods. Time integration, as well as discretization of the viscous fluxes and specification of boundary conditions, is addressed in Section 4.

2. METHODS FOR SMOOTH FLOWS

The most obvious choice to discretize the governing equations in smooth parts of the flow field is high-order approximations of the derivative operators that appear in Equation 3. Straightforward FD semidiscretization of the model equation (Equation 4) on a grid with uniform spacing b and nodes $x_j = j \cdot b$ yields

$$\frac{dv_j}{dt} = -Df_j, \quad (10)$$

where $v_j(t) \approx u(x_j, t)$ and $Df_j \approx \partial f / \partial x(x_j, t)$.

A class of explicit and implicit (compact) approximations for Df_j was devised by Lele (1992),

$$\sum_{m=-M}^M a_m Df_{j+m} = \frac{1}{b} \sum_{\ell=-L}^L b_\ell f_{j+\ell}, \quad (11)$$

where $f_j = f(v_j(t))$, and appropriate values of the coefficients a_m , b_ℓ , are determined so as to maximize formal accuracy [the minimum attainable truncation error being $O(b^{2(L+M)})$] or to shape the spectral response of the resulting scheme and improve the representation of the Fourier modes with the highest wave numbers supported by the computational mesh (Vichnevetsky & Bowles 1982). The latter idea has made the so-called DRP (dispersion-relation-preserving) schemes flourish (Bogey & Bailly 2004, Lui & Lele 2002, Tam & Webb 1993, Zingg et al. 1993). Central difference approximations, recovered for $b_{-\ell} = -b_\ell$, $a_{-m} = a_m$, have null dissipation

error in a linear setting (Lele 1992) and are therefore ideal candidates for computations free of spurious numerical diffusivity.

In the context of compressible flows, it is useful to cast the approximation given in Equation 10 in local discrete conservation form; i.e., one looks for a numerical flux $\hat{f}_{j+1/2}$, such that

$$Df_j = \frac{1}{b}(\hat{f}_{j+1/2} - \hat{f}_{j-1/2}). \quad (12)$$

For linear representations of the first derivative of the type given in Equation 11, explicit formulas for the corresponding numerical fluxes are easily found:

$$\sum_{m=-M}^M a_m \hat{f}_{j+1/2+m} = \sum_{\ell=-L+1}^L c_\ell f_{j+\ell}, \quad (13)$$

where $c_{-L+1} = -a_{-L}$, $c_\ell = c_{\ell-1} - a_{\ell-1}$, and $\ell = -L+2, \dots, L$. Explicit (even nonlinear) approximations of the numerical flux of the type $\hat{f}_{j+1/2} = \hat{f}(v_{j-L+1}, \dots, v_{j+L})$, which satisfy the consistency condition $\hat{f}(v, \dots, v) = f(v)$, guarantee global discrete conservation of the integral of u for the scalar model equation (and of total mass, momentum, and energy for the Euler equations) through the telescopic property and satisfy the hypotheses of the Lax-Wendroff theorem (Lax & Wendroff 1960, LeVeque 1990), which guarantees convergence to weak solutions, provided the approximation itself converges.

Central derivative approximations have been widely used in the literature, especially for wave propagation problems in which nonlinearities are weak (Colonius & Lele 2004). However, it is known that application of standard central discretizations to high-Reynolds number fluid turbulence typically leads to numerical instability, owing to the accumulation of the aliasing errors resulting from discrete evaluation of the nonlinear convective terms (Phillips 1959). Such deficiency can also be traced back to the failure to discretely preserve quadratic invariants associated with the conservation equations (Lilly 1965, Orlandi 2000). Although finite viscosity may help to stabilize calculations, it is usually safer to revert to alternative discretization techniques capable of ensuring stability in the inviscid limit.

2.1. Upwinding

The upwinding approach, commonly followed in the gasdynamics community, is based on the idea that solutions of the Euler equations propagate along characteristics, and therefore a stable numerical method should also propagate its information in the same characteristic direction (Moretti 1979). The standard practice in the FD framework, referred to as flux vector splitting (Steger & Warming 1981), is to split the flux function into positive and negative parts $f(u) = f^+(u) + f^-(u)$, associated with non-negative and nonpositive propagation velocities, respectively, i.e., $df^+/du \geq 0$, $df^-/du \leq 0$, which are discretized by means of left- and right-biased approximations, respectively, thus ensuring linear stability.

In the finite-volume (FV) framework, upwinding is usually achieved through the flux difference splitting (or Godunov) approach. Considering an FV semidiscretization of Equation 4,

$$\frac{d\bar{v}_j}{dt} = -\frac{1}{b}(f(v_{j+1/2}) - f(v_{j-1/2})), \quad (14)$$

where $\bar{v}_j(t) = 1/b \int_{x_{j-1/2}}^{x_{j+1/2}} v(x, t) dx$ is the spatial average of the approximate solution over the cell $I_j = (x_{j-1/2}, x_{j+1/2})$, a suitable reconstruction operator is used to determine approximate left and right states at the cell interfaces, $v_{j+1/2}^\pm$, and the interface flux $f(v_{j+1/2})$ is replaced with the numerical flux resulting from an exact (or approximate) Riemann solver (Roe 1981, Toro 2009), formally

Local discrete conservation:

property of a numerical scheme that allows the definition of a numerical flux

Global discrete conservation:

discrete counterpart of physical integral conservation properties

FV: finite volume

ADER: arbitrary accuracy schemes based on derivative Riemann problems

Numerical dissipation: nonphysical dissipation introduced to stabilize shock calculations

$\hat{f}_{j+1/2} = \mathcal{R}(v_{j+1/2}^-, v_{j+1/2}^+)$. For example, Roe's (1981) approximate Riemann solver dictates

$$\mathcal{R}(v_{j+1/2}^-, v_{j+1/2}^+) = \frac{1}{2}(f(v_{j+1/2}^-) + f(v_{j+1/2}^+)) - \frac{|a_{j+1/2}|}{2}(v_{j+1/2}^+ - v_{j+1/2}^-), \quad (15)$$

where $a_{j+1/2} = (f(v_{j+1/2}^+) - f(v_{j+1/2}^-)) / (v_{j+1/2}^+ - v_{j+1/2}^-)$ is a characteristic speed associated with the intermediate state. In Godunov's (1959) original formulation, piece-wise constant reconstructions were assumed, leading to the general first-order flux

$$\hat{f}_{j+1/2} = \mathcal{R}(\bar{v}_j, \bar{v}_{j+1}). \quad (16)$$

The extension to a higher order of accuracy is achieved by replacing piece-wise constant reconstructions with piece-wise polynomial reconstructions (van Leer 1979). In the ADER (arbitrary accuracy schemes based on derivative Riemann problems) approach (Toro et al. 2001), the numerical flux is evaluated by solving the generalized Riemann problem using a semianalytical method. The approximate solution is given as a Taylor time expansion at the cell interface position, thus allowing the construction of one-step schemes with arbitrary accuracy in both space and time.

Upwinding has the main effect of damping the Fourier modes with the highest supported wave numbers, with a subsequent stabilizing effect on the numerical solution. High-order upwind schemes have often been used for DNS of shock-free compressible turbulence (Foyi et al. 2004, Rai & Moin 1993, Pirozzoli et al. 2008), with a good degree of success. However, as pointed out by Park et al. (2004), the numerical dissipation introduced by upwinding can be harmful for LES, for which proper resolution of marginally resolved wave numbers is crucial, as it may hamper the effect of subgrid-scale models.

2.2. Filtering

Filtering the computed solution is a commonly used practice to cure nonlinear instabilities of central schemes, while retaining high-order accuracy. High-order filters were introduced by Lele (1992) and used by Visbal & Gaitonde (1999, 2002) for the solution of conservation laws on stretched, curvilinear, and deforming meshes. The latter authors considered compact filters of the form

$$\alpha \hat{v}_{j-1} + \hat{v}_j + \alpha \hat{v}_{j+1} = \sum_{\ell=0}^L \frac{d_\ell}{2} (v_{j+\ell} + v_{j-\ell}), \quad (17)$$

where the coefficients d_ℓ on the right-hand side are constrained to be functions of the free parameter α through Taylor- and Fourier-series analyses, so as to achieve high-order accuracy, and selectively damp the supported Fourier modes with the highest wave numbers. A selective filtering strategy was developed by Bogey & Bailly (2004) based on optimized explicit filters of the type given in Equation 17 with $\alpha = 0$ and $L = 4, 5, 6$. The authors also suggested a possible use of the selective filtering to model the effects of the unresolved (subgrid) flow scales, replacing classical eddy-viscosity models. As for upwinding, filtering can eliminate spurious instabilities, but it implies artificial energy draining that adds to the physical energy draining caused by molecular viscosity. More accurate control over diffusive effects is achieved by reverting to numerical methods capable of conserving energy in the absence of viscosity.

2.3. Energy-Consistent Schemes

Several attempts have been made to design nonlinearly stable numerical schemes by replicating the energy-preservation properties of the governing equations in the discrete sense. Most efforts

are based on the idea of splitting the convective derivatives found in Equation 3 as either

$$\frac{\partial \rho u_i \varphi}{\partial x_i} = \frac{1}{2} \frac{\partial \rho u_i \varphi}{\partial x_i} + \frac{1}{2} \varphi \frac{\partial \rho u_i}{\partial x_i} + \frac{1}{2} \rho u_i \frac{\partial \varphi}{\partial x_i}, \quad (18)$$

as proposed by Feiereisen et al. (1981), or

$$\frac{\partial \rho u_i \varphi}{\partial x_i} = \frac{1}{2} \frac{\partial \rho u_i \varphi}{\partial x_i} + \frac{1}{2} u_i \frac{\partial \rho \varphi}{\partial x_i} + \frac{1}{2} \rho \varphi \frac{\partial u_i}{\partial x_i}, \quad (19)$$

as proposed by Blaisdell et al. (1996), where φ stands for a generic transported scalar property, being unity for the continuity equation, u_j for the momentum equation, and H for the total energy equation. Discretization of the mass and momentum equations in the split form given in Equation 18 (often improperly referred to as skew-symmetric) implies kinetic energy preservation at the semidiscrete level (Honein & Moin 2004), provided the difference operators satisfy the summation-by-parts property (Strand 1994) that automatically holds in the case of periodic problems for central (both explicit and implicit) approximations of any order (Mansour et al. 1979). From another viewpoint, discretization of Equation 19 guarantees minimization of the aliasing error (Blaisdell et al. 1996). Additional robustness in the presence of strong density variations is gained (Kennedy & Gruber 2008) by expanding the convective derivatives in the generalized form

$$\begin{aligned} \frac{\partial \rho u_i \varphi}{\partial x_i} = & \alpha \frac{\partial \rho u_i \varphi}{\partial x_i} + \beta \left(u_i \frac{\partial \rho \varphi}{\partial x_i} + \rho \frac{\partial u_i \varphi}{\partial x_i} + \varphi \frac{\partial \rho u_i}{\partial x_i} \right) \\ & + (1 - \alpha - 2\beta) \left(\rho u_i \frac{\partial \varphi}{\partial x_i} + \rho \varphi \frac{\partial u_i}{\partial x_i} + u_i \varphi \frac{\partial \rho}{\partial x_i} \right). \end{aligned} \quad (20)$$

This arrangement leads to semidiscrete energy conservation in the case $\alpha = \beta = 1/4$. Ducros et al. (2000) showed that the split convective forms of Equations 18 and 19 give rise to locally conservative schemes, when the derivative operators are discretized with explicit central formulas (with order up to six). Pirozzoli (2010) has recently proved that this holds true for explicit central formulas of arbitrary order of accuracy and presented computationally inexpensive numerical fluxes also for the split form given in Equation 20. Apparently, compact derivative approximations applied to the split form of the convective terms do not guarantee local discrete conservation. For a sound mathematical formulation of convective splitting of the compressible Euler equations that satisfies the mathematical conditions required for skew symmetry, the reader may refer to Morinishi (2010).

2.4. Entropy-Consistent Schemes

Stabilization of numerical schemes can also be achieved by enforcing, at the discrete level, the conservation properties associated with entropy. Tadmor (1987) developed a second-order FV, locally conservative discretization of the Euler equations that discretely satisfies Equation 6, resulting in the formulation of the numerical flux

$$\mathbf{f}_{j+1/2} = \int_0^1 \mathbf{f}(\mathbf{v}_j + \xi(\mathbf{v}_{j+1} - \mathbf{v}_j)) d\xi, \quad (21)$$

which was also considered by Jameson (2008b). Note that Equation 21 is rather expensive, as it requires the evaluation of the entropy variables associated with the left and right states, and the use of quadrature formulas to approximate the integral. More general expressions for the numerical flux of entropy-consistent schemes can be found in Tadmor (2003).

An alternative strategy to ensure entropy stability was proposed by Gerritsen & Olsson (1996), who split the flux vector into conservative and nonconservative parts, exploiting the homogeneity

Split convective form: arrangement of convective derivative terms aimed at enforcing discrete preservation of quadratic invariants or minimizing the aliasing error

SHOEC: split high-order entropy conserving

SSP: strong stability preserving

property given in Equation 9:

$$\frac{\partial \mathbf{f}}{\partial x} = \frac{\beta}{\beta + 1} \frac{\partial \mathbf{f}}{\partial x} + \frac{1}{\beta + 1} \mathbf{B} \frac{\partial \mathbf{v}}{\partial x}. \quad (22)$$

This splitting, referred to in the literature as canonical or entropy splitting, leads to a discrete analog of the energy estimate given in Equation 8 (with consequent benefits in terms of nonlinear stability) if the first derivatives in Equation 22 are replaced with summation-by-parts operators. The resulting schemes are referred to as split high-order entropy conserving (SHOEC) because discrete entropy conservation in the sense of Equation 6 follows (Tadmor 1984). The performance of SHOEC schemes is apparently affected by the choice of the splitting parameter β , and Sandham et al. (2002) suggested that a value of $\beta \approx 4$ is appropriate for numerical simulation of compressible turbulence, whereas $\beta \approx 2$ is more appropriate for numerical simulation of isotropic turbulence at zero viscosity (Honein & Moin 2004). As is the case for all formulations of the Euler equations that rely on the use of entropy variables, SHOEC schemes suffer from reduced computational efficiency, owing to the swap from conservative to entropy variables, which requires CPU-intensive transcendental operations.

Honein & Moin (2004) developed entropy-consistent schemes by applying the convective splitting given in Equation 18 to the Euler equations, upon replacement of the total energy equation with the entropy equation. This approach effectively preserves the integrals of ρs and ρs^2 . Those authors also showed that the internal and total energy equations can be manipulated in such a way that the split convective form of the entropy equation automatically follows. This is particularly advantageous, as the total energy equation is typically used in compressible flow codes, especially for shock calculations.

2.5. Numerical Tests

The robustness of the numerical methods considered above can be appreciated from numerical simulations of isotropic compressible turbulence at zero viscosity on a 32^3 grid, as proposed by Honein & Moin (2004). The calculations are initialized with a field of synthetic isotropic solenoidal turbulence with an assigned wave-number spectrum $E(k) \sim (k/k_0)^4 \exp[-2(k/k_0)^2]$, peaking at $k_0 = 6$, and fluctuations of temperature and density are initially set to zero. Time integration is performed by means of a third-order strong-stability-preserving (SSP) Runge-Kutta algorithm (Shu & Osher 1989) (see Section 4.2). This computational arrangement causes energy to be quickly transferred to the smallest scales, posing a significant challenge to numerical algorithms, and leading to nonlinear instability in the absence of aliasing control mechanisms.

The results of numerical simulations are reported in **Figure 1** for initial turbulent Mach numbers $M_{t0} = 0.07$ and 0.3. In the former case, the exact solution, computed by means of a dealiased spectral method (Honein & Moin 2004), has nearly constant total kinetic energy ($K = \int \rho u^2 dV$), constant total entropy ($S = \int \rho s dV$), and the root-mean-square density levels off to $\rho' / \rho_0 / M_{t0}^2 \approx 0.35$, after an initial transient. As seen from the figure, central discretization of the equations cast in divergence form (schemes D2 and D6) leads to divergence of the numerical solution on a timescale shorter than one eddy turnover time, regardless of the order of accuracy. Divergence is associated with an increase of kinetic energy and loss of total entropy. Upwinding (scheme D7) curbs the divergence but dissipates most of the kinetic energy by 10 eddy turnover times and leads to an entropy increase. The results of simulations performed at $M_{t0} = 0.07$ show that splitting the convective terms (schemes S6 and K6) helps conserve kinetic energy and entropy, although spurious growth of density fluctuations is observed. This is also true of the Tadmor entropy-conserving scheme (scheme T2) that exhibits larger errors over long times,

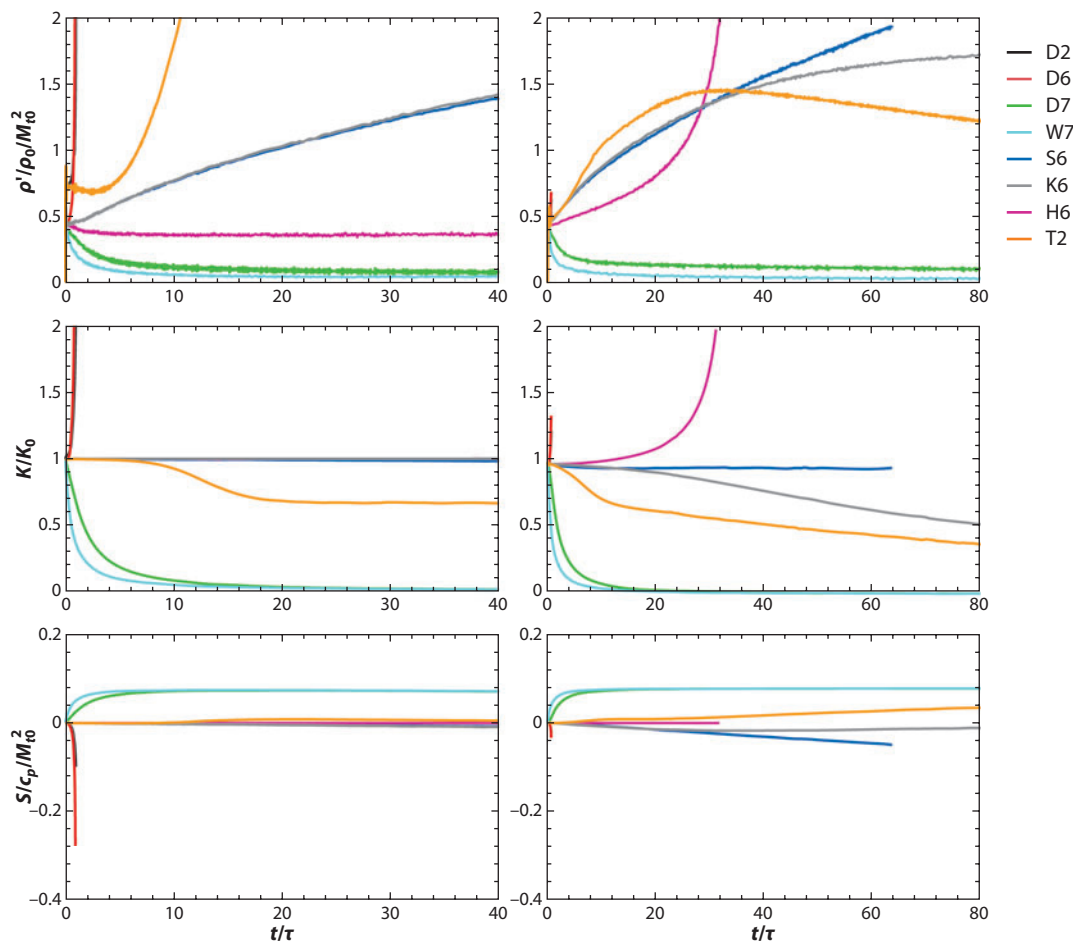


Figure 1

Simulations of isotropic turbulence at zero viscosity at $M_{t0} = 0.07$ (left column) and $M_{t0} = 0.3$ (right column); time evolution of root-mean-square density fluctuations, total kinetic energy, and total entropy (τ is the eddy turnover time). Dn are the n -th order discretization of the divergence form, W7 is the seventh-order weighted essentially nonoscillatory (WENO) scheme, S6 is the sixth-order discretization of the convective split form given in Equation 19, K6 is the sixth-order discretization of the convective split form given in Equation 20, H6 is the sixth-order entropy-consistent scheme of Honein & Moin (2004), and T2 is the second-order entropy-conserving scheme of Tadmor (1984) ($\beta = 4$). Lines are interrupted where numerical divergence occurs.

but does not lead to divergence, however. The entropy-consistent scheme of Honein & Moin (2004) (scheme H6) succeeds in correctly predicting the limiting inviscid behavior of the system for this specific test case. The results at $M_{t0} = 0.3$ show failure of all methods at large times, except the split convective scheme based on Equation 20 (scheme K6) and the Tadmor scheme (scheme T2).

3. METHODS FOR SHOCKED FLOWS

All methods presented in the previous section suffer from spurious Gibbs oscillations near shock jumps, which may lead to nonlinear instabilities. The onset of oscillations can be avoided (or at least

Shock-capturing:
numerical procedure
to compute shock
waves without
resolving their internal
structure

limited) by following two strategies. The shock-fitting approach treats shock waves (separately from the rest of the flow) as genuine discontinuities, with dynamics governed by their own algebraic equations, and uses the Rankine-Hugoniot relations as a set of nonlinear boundary conditions to relate the states on the two sides of the discontinuity (Moretti 1987). The shock-capturing approach uses the same discretization scheme at all points and achieves regularization through the addition of numerical dissipation, which inhibits the onset of Gibbs oscillations. Although the former approach often guarantees more accurate representations of shocked flows (Johnsen et al. 2010), it is only feasible in cases in which the shock topology is extremely simple and no shock wave forms during the calculation. In this review, we discuss only the shock-capturing approach.

3.1. Classical Shock-Capturing Methods

Historically, most shock-capturing schemes have been constructed with the mathematical properties of the entropy solution of the model scalar conservation law given in Equation 4 in mind, including the following: (a) monotonicity preservation, i.e., initially monotone data remain monotone for all times (Godunov 1959); (b) total variation diminishing (TVD), i.e., solutions of Equation 4 satisfy $TV(u(\cdot, t_2)) \leq TV(u(\cdot, t_1))$, $\forall t_2 \geq t_1$, where $TV(u(\cdot, t))$ denotes the total variation of u (Harten 1983a); and (c) monotonicity, i.e., let $u(x, t)$, $v(x, t)$ be any two entropy solutions of Equation 4, with $u(x, 0) \leq v(x, 0)$, $\forall x$, and then $u(x, t) \leq v(x, t)$, $\forall x, t$ (Harten et al. 1986).

Shock-capturing schemes implementing the above conditions in the discrete sense are abundant in the literature, mainly because of the availability of theorems that guarantee convergence to weak solutions for TVD schemes (LeVeque 1990) and to the (unique) entropy solution for monotone schemes (Harten et al. 1986). However, monotone schemes are at most first-order accurate (Harten et al. 1986), which explains the wide success enjoyed by TVD schemes, especially in the 1980s. An excellent overview of the early contributions to TVD schemes is given by Yee (1989).

Classical TVD schemes were found to suffer from loss of accuracy at both smooth and non-smooth extrema, which stimulated researchers to pursue alternatives for constructing uniformly high-order accurate shock-capturing schemes. The successful family of essentially nonoscillatory (ENO) schemes (Harten et al. 1987) is based on the idea of determining the numerical flux from a high-order reconstruction over an adaptive stencil that is selected to avoid as much interpolation across discontinuities as possible, thus minimizing Gibbs oscillations. ENO reconstructions of order r guarantee that $TV(v^{n+1}) \leq TV(v^n) + O(b^r)$, which implies uniform boundedness of the total variation. ENO schemes do have problems of convergence to steady solutions and loss of accuracy, due to free stencil adaptation (Rogerson & Meiburg 1990). Shu (1990) resolved the issue by biasing toward the best central stencil in the adaptation procedure. Suresh & Huynh (1997) proposed an alternative technique to circumvent the loss of accuracy in TVD schemes, based on the enforcement of the monotonicity-preservation constraint at the discrete level.

The class of the weighted essentially nonoscillatory (WENO) schemes, first introduced by Liu et al. (1994), and generalized and improved by Jiang & Shu (1996), is based on the idea of constructing a high-order numerical flux from a convex linear combination of lower-order polynomial reconstructions over a set of staggered stencils, with weights selected to achieve maximum formal order of accuracy in smooth regions, and assigning (nearly) zero weight to reconstructions on stencils crossed by discontinuities. As illustrated in **Figure 2**, $L+1$ substencils, each including L points, $S_\ell = \{x_{j-L+\ell+1}, \dots, x_{j+\ell}\}$, $\ell = 0, \dots, L$, are considered out of the overall reconstruction stencil $S = \{x_{j-L+1}, \dots, x_{j+L}\}$, which includes a total of $2L$ points. Assuming for now $f(u) = f^+(u)$, and

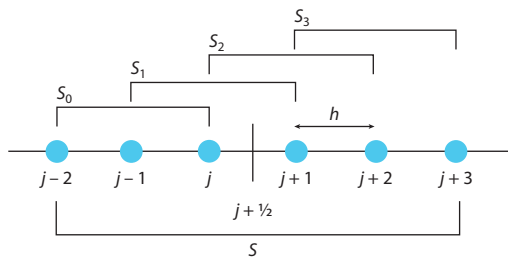


Figure 2

Definition of stencils for weighted essentially nonoscillatory (WENO) reconstruction ($L = 3$).

suppressing the superscripts, the FD WENO numerical flux is constructed as

$$\hat{f}_{j+1/2} = \sum_{\ell=0}^L \omega_{\ell} \hat{f}_{j+1/2}^{(\ell)}, \quad (23)$$

where $\hat{f}_{j+1/2}^{(\ell)}$ is the numerical flux resulting from polynomial reconstruction over the stencil S_{ℓ} ,

$$\hat{f}_{j+1/2}^{(\ell)} = \sum_{m=0}^{L-1} c_{\ell m} f_{j-L+1+\ell+m}, \quad (24)$$

and the weights, defined as

$$\omega_{\ell} = \frac{\alpha_{\ell}}{\sum_{m=0}^L \alpha_m}, \quad \alpha_{\ell} = \frac{d_{\ell}}{(\epsilon + \beta_{\ell})^2}, \quad (25)$$

are functions of the smoothness measurements associated with the substencils

$$\beta_{\ell} = \sum_{m=1}^{L-1} \left(\sum_{n=1}^{L-1} \gamma_{\ell mn} f_{j-L+1+\ell+n} \right)^2. \quad (26)$$

The key to the success of WENO reconstructions is the proper selection of the linear weights d_{ℓ} , so as to achieve maximum order of accuracy ($2L$), and the design of the nonlinear weights ω_{ℓ} , such that if the stencil S_{ℓ} contains a jump, $\omega_{\ell} \approx 0$ follows. Upwinding is achieved by suitable biasing in the reconstruction procedure. With regard to the positive part of the flux, in Jiang & Shu's (1996) original approach a left-biased reconstruction was obtained discarding the right-most substencil (i.e., $d_L = 0$), thus reducing the overall stencil width (and the formal order of accuracy) to $2L - 1$. Weirs & Candler (1997) suggested the use of a central stencil to reduce the numerical viscosity of WENO schemes and biased the reconstruction by redefining the smoothness measurement for the right-most substencil, setting $\beta_L = \max_{0 \leq \ell \leq L} \beta_{\ell}$, thus preventing the selection of the fully downwinded stencil (S_L). Formulas for the negative split flux are easily obtained by symmetry about $x_{j+1/2}$. Comprehensive compilations of WENO coefficients $c_{\ell m}$, d_{ℓ} , and $\gamma_{\ell mn}$ can be found in Weirs & Candler (1997), Balsara & Shu (2000), and Gerolymos et al. (2009).

Provable properties of WENO schemes are limited to the scalar case and include convergence for smooth solutions (Jiang & Shu 1996) and the existence and stability of discrete shock profiles for the third-order version of the scheme (Jiang & Yu 1996). Nevertheless, WENO schemes are extremely robust in practice and have been applied successfully (both in their FD and FV versions) to a wide range of physical problems. For an overview of the wide-ranging ramifications of the WENO idea, the reader may consult the review paper by Shu (2009).

Many investigators have tried to improve the properties of WENO schemes, especially in terms of their behavior in smooth regions. Most research (Weirs & Candler 1997) has been

WENO: weighted essentially nonoscillatory

Hybrid scheme:

method based on replacement of a nondissipative baseline scheme with a shock-capturing scheme near shocks

Shock sensor:

indicator for preliminary identification of critical (shocked) cells

focused on the attempt to modify the linear weights and the polynomial reconstruction coefficients to improve the behavior in Fourier space and extend the range of well-resolved wave numbers, in the spirit of the DRP approach. Follow-up studies of the optimized WENO strategy have been reported by Hill & Pullin (2004), Martín et al. (2006), and Taylor et al. (2007). An alternative approach to conjugate the spectral-like resolution properties of compact schemes with the shock-capturing properties of WENO was undertaken by Deng & Zhang (2000), who developed weighted compact nonlinear schemes by coupling WENO interpolation of conservative variables to the intermediate nodes and compact conservative approximation of the flux derivatives at the grid nodes. In the author's experience, WENO schemes and their variants are highly accurate and quite robust. However, they are computationally expensive, mostly because of the large number of floating point operations required for the evaluation of the smoothness measurements. Furthermore, spectral-like resolution is never achieved owing to the noxious effect of the nonlinear weights, unless the weights are locally frozen in smooth regions, as suggested by Hill & Pullin (2004) and Taylor et al. (2007).

The performance of WENO schemes is illustrated in **Figure 3**, which shows the numerical simulation results of Lax & Liu's (1998) test case 13. The solution of the 2D Riemann problem involves a complex shock diffraction pattern leading to the formation of a Mach stem and Kelvin-Helmholtz roll-up of the vertical slip line. **Figure 3** shows that both the TVD and the WENO methods robustly capture the shock-wave pattern, without significant spurious oscillations. Differences are observed in the predicted structure of the slip line, which exhibits finer-scale roll-up for the WENO schemes and the formation of secondary vortex cores for the seventh-order version.

3.2. Hybrid Schemes and Nonlinear Filtering

This class of methods is based on the idea of endowing a baseline spectral-like scheme with shock-capturing capability through local replacement with a classical shock-capturing scheme or through the controlled addition of the dissipative part of a shock-capturing scheme, which is made to act as a nonlinear filter. A key role in this class of schemes is played by shock sensors that must be defined in such a way that numerical dissipation is effectively confined in shocked regions, so that it does not pollute smooth parts of the flow field.

Lee et al. (1997) used a hybrid discretization comprising a sixth-order compact scheme and a sixth-order shock-capturing ENO scheme to analyze the interaction of a (nominally) planar shock wave with a field of isotropic turbulence. As the shock location in that configuration is approximately known, the switch between the two schemes was established a priori, and ENO was only activated in the shock-normal direction, over 12 points upstream and 12 points downstream of the mean shock location. Adams & Shariff (1996) first considered a truly adaptive hybrid discretization, consisting of a baseline compact upwind scheme coupled with a fifth-order ENO scheme. A simple switch, based on the local gradient of the flux vector components, was used to mark critical cells, which were subsequently padded with buffer cells on each side to prevent the onset of oscillations arising from the coupling of schemes with different properties. The method was expanded by Pirozzoli (2002), who developed a fully conservative formulation by hybridizing a fifth-order compact upwind numerical flux (of the form of Equation 13) with a seventh-order WENO flux, the switch being based on the local density gradient. **Figure 3d** shows the results obtained with this method for the Lax-Liu Riemann problem. Compared to baseline WENO schemes, the hybrid formulation yields a more accurate representation of smooth flow features, with the formation of additional discrete vortices in the shear layer, but it also yields some spurious wiggles near shocks. For this specific test case, the reduction of computational cost with respect to standard seventh-order WENO is approximately 50%. Further improvements to hybrid

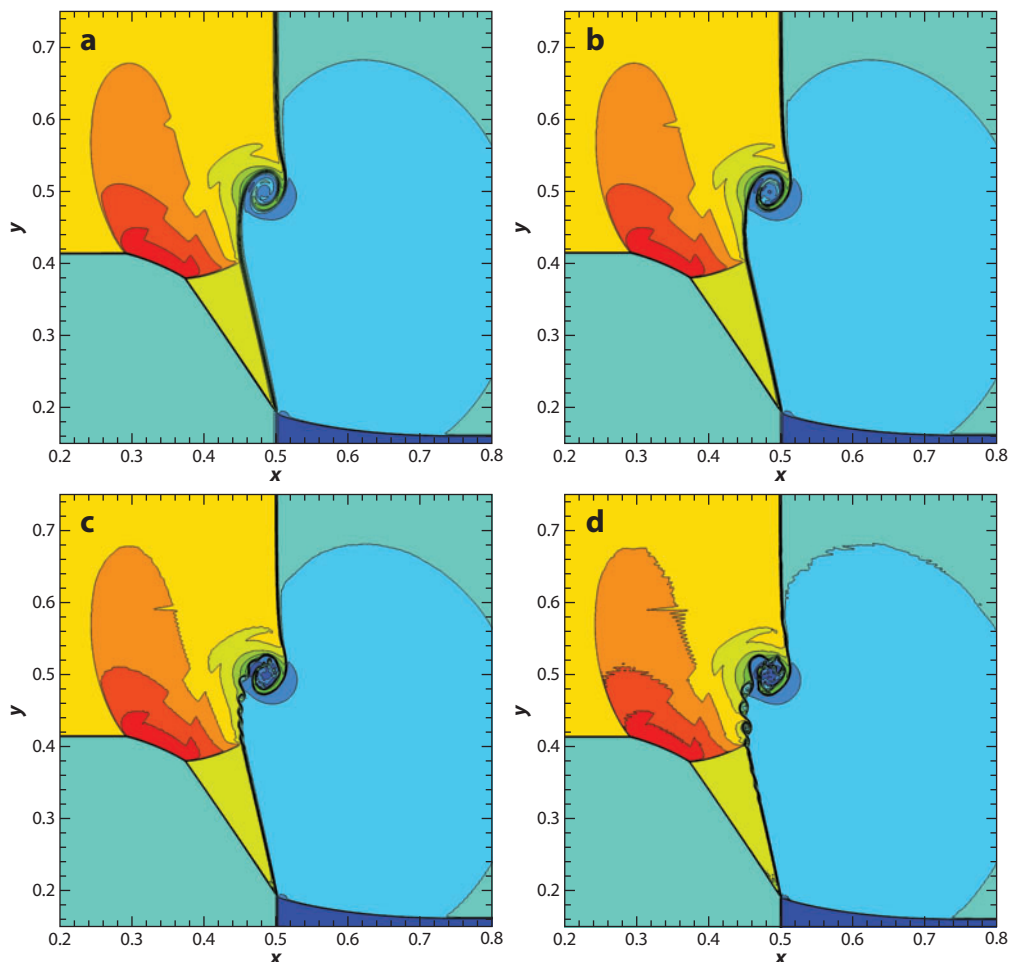


Figure 3

Numerical simulation of Lax & Liu's (1998) test case 13 on a mesh with $b_x = b_y = 1/1,200$. Only the central part of the domain is shown. Time integration is performed with a third-order strong-stability-preserving (SSP) Runge-Kutta algorithm (the Courant number was set to 0.6). Fourteen equally spaced density contours are shown, from 0.6 to 2.3. (a) Second-order total variation diminishing (TVD) scheme with a Van Leer limiter. (b) Fifth-order weighted essentially nonoscillatory (WENO) scheme. (c) Seventh-order WENO scheme. (d) Hybrid compact/WENO scheme (Pirozzoli 2002). In panel d, WENO is activated at the intermediate node $x_{j+1/2}$ (and two neighboring nodes to the left and right of it), whenever $|\rho_{j+1} - \rho_j| \geq 0.2$.

WENO schemes were introduced by Ren et al. (2003) and Hill & Pullin (2004). In particular, the latter authors used a baseline central DRP-like discretization (tuned for LES of compressible turbulence) and designed a shock sensor based on the ratio of the largest to the smallest WENO smoothness indicators, $\lambda = (\max_\ell \beta_\ell) / (\min_\ell \beta_\ell + \varepsilon)$, that becomes large whenever one or more WENO substencils are crossed by discontinuities.

Although hybrid schemes are now frequently used, it appears that the issues related to coupling of the underlying schemes have not been thoroughly investigated. A notable exception is the work of Larsson & Gustafsson (2008), who applied Kreiss stability theory to the analysis of hybrid methods. Interestingly, they found that the coupling of two separately stable schemes, in some

instances, may give rise to an unstable system. However, if either scheme is dissipative, the coupled system is strongly stable, thus endorsing the use of upwind shock-capturing schemes around discontinuities. Hybrid schemes can also be designed to have a smooth transition between two different schemes, through the definition of suitable blending functions (e.g., Ren et al. 2003). In the author's experience, this strategy does not add much to the stability and accuracy of the resulting algorithm (see also Taylor et al. 2007).

The nonlinear filtering approach was introduced by Yee et al. (1999), who designed a low-dissipative, shock-capturing algorithm based on SHOEC discretization, augmented with the artificial dissipation of a TVD scheme. With regard to the scalar conservation law, the authors proposed the following semidiscretization:

$$\frac{dv_j}{dt} = -Df_j - \frac{1}{b}(f_{j+1/2}^* - f_{j-1/2}^*), \quad (27)$$

where the filter numerical flux is defined as

$$f_{j+1/2}^* = k\Psi_{j+1/2}d_{j+1/2}. \quad (28)$$

Here k is a (problem-dependent) constant, $d_{j+1/2}$ is the artificial flux of a second-order TVD scheme, and $\Psi_{j+1/2}$ is the Harten (1978) switch, which is designed to be near unity near shocks and near zero in smooth parts of the flow. As a consequence, the algorithm automatically localizes numerical dissipation around discontinuities. The extension of the methodology to the multidimensional Euler equations involves application of the nonlinear filter to each characteristic field; hence the frequently used terminology of characteristic-based filters is also given to this class of schemes. An advantage of the nonlinear filtering approach over hybrid methods is that, in the case of explicit multistage time integration, the filter numerical flux can be applied at the end of the full time step, rather than at each substep, with substantial savings in terms of CPU time (but with some loss in terms of robustness). Ducros et al. (1999) modified the original nonlinear filtering approach by adopting the artificial flux of the Jameson-Schmidt-Turkel scheme (as from Equation 30), and replacing the Harten switch in Equation 28 with the product of the Jameson sensor given in Equation 32 and another shock indicator

$$\Theta = \frac{(\nabla \cdot u)^2}{(\nabla \cdot u)^2 + (\nabla \times u)^2 + \epsilon}, \quad 0 \leq \Theta \leq 1, \quad (29)$$

frequently referred to as the Ducros sensor. Further modifications of the method were introduced by Garnier et al. (2001), who replaced the TVD artificial flux with the dissipative part of ENO and WENO schemes and showed that the Ducros sensor is capable of distinguishing turbulent fluctuations from shocks better than the Harten switch. The idea of adaptively filtering the numerical solution for shock-capturing also exists in the work of Visbal & Gaitonde (2005) and Bogey et al. (2009), who developed computational strategies based on high-accuracy central discretization of spatial derivatives, supplemented with selective filtering of the computed solution. Shock-capturing is achieved by locally decreasing the filter bandwidth in critical regions.

A key ingredient in the formulation of hybrid and nonlinear filtering schemes is the proper specification of the shock sensor. The Ducros sensor represents a simple and frequently adopted choice. Alternative simple shock sensors include those proposed by Hill & Pullin (2004), Visbal & Gaitonde (2005), and Larsson et al. (2007) (the latter being a modification of the Ducros sensor). More elaborate sensors for shock waves and shears, based on multiresolution wavelet analysis, were proposed by Sjögreen & Yee (2000) and Yee & Sjögreen (2007).

The performance of shock sensors in practical computations can be seen in **Figure 4**, which illustrates results of the application of some of the indicators mentioned above to the instantaneous flow field obtained from DNS of transonic shock/boundary layer interactions (SBLIs) (Pirozzoli

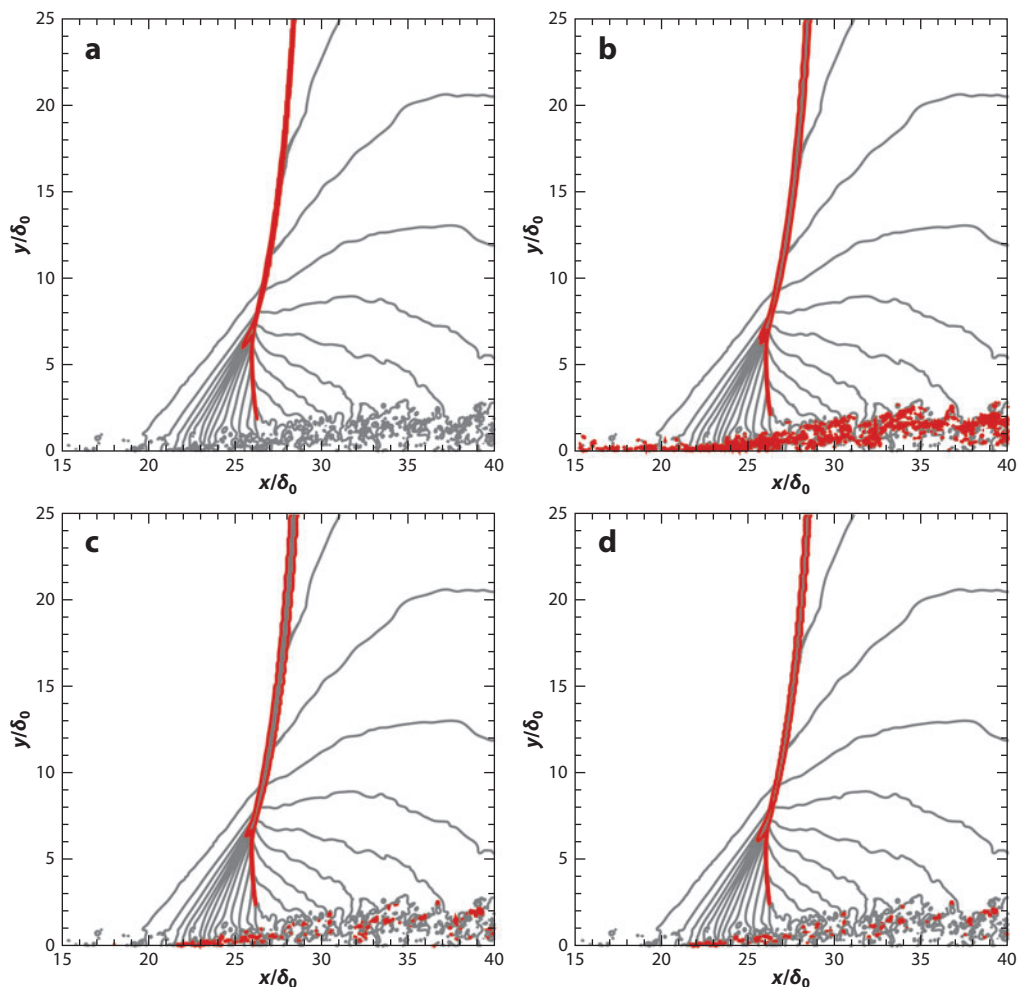


Figure 4

Performance of shock sensors for direct numerical simulation of transonic shock/boundary layer interactions. Black lines denote pressure isolines (24 equally spaced levels, $1 \leq p/p_\infty \leq 1.85$), and red lines indicate isolines of shock sensors. (a) Ducros sensor (Ducros et al. 1999). (b) Jameson sensor (Jameson et al. 1981). (c) Hill & Pullin sensor (Hill & Pullin 2004). (d) Visbal & Gaitonde sensor (Visbal & Gaitonde 2005). δ_0 denotes the incoming boundary layer thickness.

et al. 2010). The threshold values for the various sensors have been selected in such a way that a similar representation of the shock system is obtained. Note that the Ducros sensor in its original formulation does not perform properly outside of the wall layer (where $\nabla \times u \approx 0$), due to excessive sensitivity to dilatational fluctuations. The sensor can be conveniently adapted to this case by setting $\epsilon = (u_\infty/\delta_0)^2$ in Equation 29, so that it is only activated when the local dilatation becomes larger than a typical large-scale velocity gradient. Moreover, for evaluation purposes, all other sensors are based on the pressure field. The data reported in **Figure 4** suggest that the modified Ducros sensor is capable of selectively isolating genuine shocks, whereas other simple sensors also mark smooth regions populated by vortical structures as critical zones.

3.3. Artificial Viscosity Methods

The basic idea of artificial viscosity methods is to explicitly introduce the amount of numerical dissipation needed to stabilize shock computations through the addition of diffusive terms that adaptively adjust to the local regularity of the solution. In von Neumann & Richtmyer's (1950) original approach, artificial diffusive terms with viscosity coefficient $\mu^* = c_\mu \rho h^2 |\partial u / \partial x|$ (where c_μ is a tunable constant) were added to the momentum and energy equations to spread shock waves to a thickness comparable to the local mesh spacing. The artificial viscosity idea also underlies the work of Jameson et al. (1981), who designed a high-order nonoscillatory numerical flux by augmenting the numerical flux of a second-order central scheme ($\hat{f}^C$), with explicit diffusive fluxes resulting from discretization of second- and fourth-derivative operators, as follows:

$$\begin{aligned}\hat{f}_{j+1/2} &= \hat{f}_{j+1/2}^C - d_{j+1/2}, \\ d_{j+1/2} &= a_{j+1/2} [\epsilon_{j+1/2}^{(2)} (v_{j+1} - v_j) - \epsilon_{j+1/2}^{(4)} (v_{j+2} - 3v_{j+1} + 3v_j - v_{j-1})],\end{aligned}\quad (30)$$

where in the case of the Euler equations the characteristic speed $a_{j+1/2}$ is replaced with the spectral radius of the Jacobian matrix of the inviscid flux at Roe's (1981) average state. The diffusion coefficients, defined as

$$\epsilon_{j+1/2}^{(2)} = k^{(2)} \psi_{j+1/2}, \quad \epsilon_{j+1/2}^{(4)} = \max(0, k^{(4)} - \epsilon_{j+1/2}^{(2)}), \quad (31)$$

are functions of the local regularity of the solution through

$$\psi_{j+1/2} = \max(\psi_j, \psi_{j+1}), \quad \psi_j = \frac{|v_{j+1} - 2v_j + v_{j-1}|}{|v_{j+1} + 2v_j + v_{j-1}|}, \quad 0 \leq \psi_j \leq 1, \quad (32)$$

and $k^{(2)} \approx 1$, $k^{(4)} \approx 0.01$ – 0.05 . The Jameson-Schmidt-Turkel artificial flux is designed in such a way that second-order dissipation is activated only near shocks, whereas fourth-order dissipation only acts in smooth regions to suppress high-frequency oscillations. In the case of the Euler equations, regularity of the solution is gauged by applying Equation 32 to the pressure field.

Von Neumann's original idea was extended by Tadmor (1989) and Guo et al. (2001), who introduced a spectrally accurate vanishing viscosity to augment the approximation of nonlinear conservation laws in spectral space. It was shown that a suitable choice of the viscosity kernel in Fourier space recovers spectral convergence for weak solutions of Burgers equation. Cook & Cabot (2004, 2005) carried over the idea of selectively damping the highest-resolved wave numbers to physical space, through the addition of an artificial stress tensor,

$$\sigma_{ij}^* = 2\mu^* S_{ij} + (\beta^* - 2/3 \mu^*) u_{k,k} \delta_{ij}, \quad (33)$$

where $S_{ij} = 1/2(u_{i,j} + u_{j,i})$. The artificial shear and bulk viscosity coefficients are defined as

$$\mu^* = c_\mu^r \eta_r, \quad \beta^* = c_\beta^r \eta_r, \quad \eta_r = \rho \Delta^{2(r+1)} |\widehat{\nabla^{2r} S}|, \quad (34)$$

where c_μ^r, c_β^r are user-specified constants, Δ is a measure of the local mesh spacing, $S = (2S_{ij} S_{ij})^{1/2}$, ∇^{2r} denotes the polyharmonic operator (i.e., the r -th power of the Laplace operator), and the caret denotes a truncated Gaussian filter, designed to have smooth η_r . A set of coefficients used by most investigators is $r = 2$, $c_\beta^2 = 1$, and $c_\mu^2 = 0.002$. Compared to the von Neumann formulation (which is recovered for $r = 0$), the choice given in Equation 34 imparts on the numerical viscosity a higher wave-number bias. Inclusion of the bulk viscosity term is the key to capturing shock waves without affecting the vorticity field, whereas the shear viscosity has the effect of a Smagorinsky-like subgrid-scale model for unresolved vortical fluctuations. As observed by Mani et al. (2009), the definition given in Equation 34 introduces unnecessary damping into resolved dilatational motions (acoustic waves). To remove such deficiency, Mani et al. (2009) replaced S with $u_{k,k}$ in Equation 34, in the definition of c_β^r , and set c_β^r to zero whenever $u_{k,k} > 0$ (i.e., in expansion

zones). Bhagatwala & Lele (2009) introduced additional selectivity in the definition of c_{β}^r , so that it effectively becomes zero in the presence of resolved acoustic motions.

Further improvements of the artificial viscosity method have led to the inclusion of the effects of strong temperature gradients (contact discontinuities), as well as species diffusion in multicomponent mixtures (Fiorina & Lele 2007, Cook 2007). Kawai & Lele (2007) extended the method to generalized curvilinear coordinates.

3.4. Shock-Capturing Through Subgrid-Scale Models

Methods of this class are based on the attempt to regularize weak solutions of the conservation equations through the addition of subgrid-scale models that drain energy from the unresolved scales of motion, in analogy to what is done for LES of smooth flows. This idea is related to the artificial viscosity method and was elaborated by Adams & Stolz (2002). The authors developed a method based on explicit filtering of the flow variables, whereby the closure terms are derived from an approximate deconvolution of the flow scales represented on the computational grid, and entropy regularization is achieved by the addition of a relaxation term driven by the amount of energy found at the smallest resolved scales. Referring to the scalar conservation law given in Equation 4, Adams & Stolz's (2002) method can be formulated as an evolution equation,

$$\frac{\partial \hat{u}}{\partial t} + \frac{\partial f(\hat{u})}{\partial x} = \mathcal{G}_1 + \mathcal{R}, \quad (35)$$

for the filtered variable $\hat{u} = G \star u$, where G is a (primary) low-pass filtering operator. The term $\mathcal{G}_1$ accounts for the effect of the computed flow scales (i.e., those represented on the computational grid) on the resolved ones (i.e., on the filtered field) and is approximated as

$$\mathcal{G}_1 = \frac{\partial f(\hat{u})}{\partial x} - \frac{\partial f(\tilde{u})}{\partial x}, \quad (36)$$

where $\tilde{u} = Q \star \hat{u}$, Q being a regularized approximation of the exact deconvolution operator G^{-1} . The effect of the interaction between the nonrepresented scales and the resolved scales is further modeled through the relaxation term

$$\mathcal{R} = -\chi(\hat{u} - G_2 \star \hat{u}), \quad (37)$$

where G_2 is a secondary (more selective) low-pass filter, and χ is an adjustable relaxation constant. The modified Equation 34 is then discretized through high-order compact differencing. A weak point of the formulation is the specification of a suitable value for the relaxation constant χ , which apparently cannot be established a priori. A dynamic procedure to estimate χ as a function of space and time was described by von Kaenel et al. (2004).

3.5. Properties of Shock-Capturing Schemes

The analysis of shock-capturing schemes is made difficult by their inherent nonlinearity, which prevents the straightforward application of the tools extensively used to evaluate linear schemes, especially Fourier analysis (Vichnevetsky & Bowles 1982). Nonlinearity also makes comparison of the relative quality of different methods, other than a case-by-case basis, almost impossible. Often, classical shock-capturing schemes have linear counterparts that are used as guidance to improve their spectral behavior. For example, freezing the nonlinear weights turns WENO schemes into linear upwind schemes, whose spectral transfer function has been used in the development of the optimized-WENO schemes. However, numerical tests show that the nonlinear features have a dramatic impact on the performance of numerical schemes, and their actual behavior may be quite

different from linear predictions. For instance, **Figure 1** shows a substantially poorer performance (in terms of increased numerical dissipation) of the seventh-order WENO scheme (scheme W7) compared with its linearized seventh-order upwind version (scheme D7). For a discussion of the impact of nonlinearities on the behavior of shock-capturing schemes in Fourier space, readers may refer to Pirozzoli (2006). Useful information related to the performance of shock-capturing schemes for shock-free turbulent flows can be found in Garnier et al. (1999) and Larsson et al. (2007).

A major flaw of shock-capturing schemes, often disregarded, is the reduction of accuracy for near shocks. Indeed, even (nominally) uniformly high-order schemes yield first-order accurate solutions downstream of moving shocks, mainly because shocks have zero measure, and therefore their location is only known to $O(b)$ on a finite grid. The loss of accuracy was highlighted in model scalar shock-sound interaction problems (Casper & Carpenter 1998, Engquist & Sjögreen 1998), and it is likely to be the cause of the slow convergence of apparently simple shock/turbulence interaction calculations (Larsson & Lele 2009). For an interpretation of the pathology in terms of numerical viscosity, the reader is referred to Efraimsson & Kreiss (1999). Shock-capturing schemes applied to systems of conservation laws also suffer from spurious postshock oscillations, especially in the case of slowly moving shock (Arora & Roe 1997), which may prevent the accurate prediction of shock/sound and shock/turbulence interactions, as observed by Johnsen et al. (2010). These pathologies are apparently unavoidable, unless one reverts to special techniques, such as subcell resolution (Harten 1989) or even to shock-fitting.

4. FURTHER TOPICS

All methods presented above are focused on the discretization of the spatial derivatives that appear in the governing equations. This section briefly addresses other issues, related to time integration and specification of numerical boundary conditions, as well as the extension to complex geometries.

4.1. Complex Geometries

It is easy to extend FD schemes to multidimensional systems of conservation laws in Cartesian domains through component-by-component (or characteristic-wise), line-wise application of 1D algorithms. Relatively complex geometries, which can be smoothly mapped to a Cartesian mesh, can also be handled by exploiting the strong conservation form of the equations (Vinokur 1974), which allows the extension of locally conservative approximations derived for uniformly spaced Cartesian meshes, thus guaranteeing convergence to weak solutions. The immersed boundary approach (Mittal & Iaccarino 2005) promises simple handling of complex geometries in the FD framework, representing a viable alternative to the adoption of body-fitted grids. However, relatively few applications of the method to compressible flows have appeared (de Tullio et al. 2007, Ghias et al. 2007, Hu et al. 2006).

FV methods provide greater flexibility than FD methods in dealing with complex geometries, as locally conservative FV schemes can be easily designed for both structured and unstructured meshes. The FV framework also allows the design of one-step methods in time (as opposed to multistage Runge-Kutta time integration commonly used for FD methods; see Section 4.2), having high-order accuracy in both space and time, which makes ADER schemes competitive with typical shock-capturing FD schemes having comparable accuracy (Titarev & Toro 2005). Furthermore, it appears that an FV formulation, with the use of suitable positivity-preserving approximate Riemann solvers, is necessary in some instances, such as the computation of compressible multicomponent flows, to avoid oscillations in the presence of material interfaces (Johnsen & Colonius 2006). With regard to FV methods for structured meshes, it is known that straightforward

dimensional splitting gives rise to second-order errors (regardless of the accuracy of the underlying reconstructions), unless computationally expensive quadratures are used to evaluate the flux integrals transverse to the direction being reconstructed. High-order accurate, quadrature-based FV schemes have been developed both in the classical method-of-lines approach with standard Riemann solvers (Titarev & Toro 2004) and in the one-step ADER framework (Titarev & Toro 2005). However, as observed by Ducros et al. (2000) and Gerolymos et al. (2009), the splitting error is usually small, and line-wise application of 1D reconstructions is quite successful in practice, yielding an accurate representation of smooth flow features and good shock-capturing properties. Unstructured meshes mandate the use of both high-order flux quadratures at cell interfaces and genuinely multidimensional reconstructions (Hu & Shu 1999), thus making high-order FV schemes highly expensive. An important step in the direction of improving the computational efficiency of high-order FV schemes was accomplished by Dumbser et al. (2007), who succeeded in designing one-step nonoscillatory FV schemes for unstructured tetrahedral meshes with arbitrary order of accuracy, without the need of quadratures. Their strategy exploits a characteristic WENO reconstruction yielding the whole polynomial information in each cell and a Cauchy-Kovalevski procedure to provide a space-time Taylor series for the conserved quantities and the physical fluxes. This information is used to construct highly accurate upwind numerical fluxes, which are subsequently integrated analytically in space and time. A comprehensive review of modern FV methods is given by Toro (2009).

Alternative strategies to achieve uniform high accuracy for flows in complex geometries, especially in an unstructured mesh framework, include spectral volume and discontinuous Galerkin (DG) methods (Wang 2007). However, these methods require the storage of several degrees of freedom per cell and therefore seem to be too demanding currently in terms of memory requirements for large-scale turbulence computations. A novel approach was proposed by Dumbser et al. (2008), who developed a unified framework for the construction of fully discrete and very-high-order accurate quadrature-free one-step FV and DG schemes on unstructured tetrahedral meshes (the so-called $P_N P_M$ schemes), where the data are represented in each cell by piecewise polynomials of degree N , and the fluxes are computed using piecewise polynomials of degree $M \geq N$. The class of FV methods is then recovered for $N = 0$, whereas DG methods are recovered for $N = M$. The intermediate classes of schemes with $M > N$ and $N \neq 0$ were found to produce computationally more efficient algorithms than conventional FV and DG methods.

Some recent studies have addressed the possibility of constructing low-dissipative, nonlinearly stable FV schemes for complex geometries. Kok (2009) designed fourth-order accurate schemes suitable for smooth curvilinear meshes that locally conserve both momentum and kinetic energy in the semidiscrete sense. Jameson (2008a) and Subbareddy & Candler (2009) have developed second-order FV methods suitable for unstructured meshes that discretely preserve kinetic energy. In particular, in the latter study, fully discrete energy conservation was obtained through a density-weighted Crank-Nicholson-like time integration, and shock-capturing was incorporated in the method in the form of a TVD filter controlled by the Ducros sensor. The development of energy- and entropy-consistent methods for unstructured meshes is dealt with in the dedicated review by Perot (2011) in this same volume.

4.2. Time Integration

Time integration in the method-of-lines framework is usually performed by means of explicit Runge-Kutta multistage schemes (Kennedy et al. 2000). Special types of so-called SSP Runge-Kutta schemes are frequently used (Gottlieb et al. 2001, Shu & Osher 1989), having the remarkable property that if a semidiscretization of the scalar conservation law given in Equation 4 is stable in

a certain norm (e.g., the TV norm) with first-order Euler forward time stepping, the multistage time discretization will maintain stability in that norm, under a suitable time step restriction. It has been argued (Shu 1997) that the use of SSP Runge-Kutta time discretizations is safer for solving hyperbolic problems in the presence of shock waves. In the author's experience, and as confirmed by inspection of the literature on the subject, standard Runge-Kutta time integration will work out just as fine for most typical DNS and LES applications. Optimized low-dissipation and low-dispersion Runge-Kutta algorithms, originally introduced for computational aeroacoustics (Hu et al. 1996), especially in their low-storage implementation (Stanescu & Habashi 1998), can also be fruitfully adopted for the purpose. Recent developments on the subject are reported by Bogey & Bailly (2004), Berland et al. (2006), and Bernardini & Pirozzoli (2009).

4.3. Boundary Conditions

The correct enforcement of boundary conditions in compressible flow simulations is of utmost importance for the accuracy and stability of numerical algorithms. An exhaustive account of the subject can be found in Colonius (2004). A critical issue in the simulation of spatially evolving wall-bounded compressible flows, not covered in that review, is the specification of suitable inlet conditions, which must guarantee a fully developed turbulent state of the boundary layer at the shortest possible distance from the inflow. A frequently used technique is the rescaling-recycling procedure developed by Lund et al. (1998) and extended to the compressible case by Sagaut et al. (2004) and Xu & Martín (2004). The basic idea consists of extracting a cross-stream slice of the flow field and recycling it to the inflow, after suitable rescaling. This approach produces a realistic turbulent boundary layer within a distance of about five boundary layer thicknesses from the inflow, and it allows the control of the skin friction and the thickness of the simulated boundary layer. Alternative techniques rely on the specification of synthetic disturbances at the inflow constructed so as to mimic coherent boundary layer structures (Pamiès et al. 2009, Sandham et al. 2003) or on an attempt to synthesize quasi-deterministic disturbances having the desired first- and second-order velocity statistics (Klein et al. 2003).

4.4. Discretization of Viscous Fluxes

Discretization of the viscous terms in the compressible Navier-Stokes equations involves consecutive derivative operations. As observed by Lele (1992), Sandham et al. (2002), and Nagarajan et al. (2003), better accuracy and robustness are achieved by expanding the diffusive terms as

$$\frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial x} \right) = \mu \frac{\partial^2 u}{\partial x^2} + \frac{\partial \mu}{\partial x} \frac{\partial u}{\partial x} \quad (38)$$

and applying dedicated second-derivative discretization to the first term on the right-hand side. This treatment is justified by spectral analysis of the approximation, showing significantly improved representation of the diffusive effects at the highest-resolved wave numbers, compared to straightforward repeated application of discrete first-derivative operators, which may result in odd-even decoupling.

5. APPLICATIONS

There are numerous applications of the methods of computational gasdynamics to the analysis of flow physics. Here we only highlight recent high-fidelity simulations of flows involving the interaction of shock waves with turbulence.

The simplest setting consists of the interaction of a (nominally) normal shock wave with a field of isotropic turbulence. Canonical shock/turbulence interactions were first investigated through

DNS by Lee et al. (1997), using a hybrid compact/ENO scheme. LES of the same problem was performed by Ducros et al. (1999), by means of a characteristic-based nonlinear filtering scheme. The problem was revisited by Larsson & Lele (2009), who carried out high-resolution calculations using a hybrid central WENO discretization with convective splitting.

Substantial efforts in the past few decades have been devoted to the analysis of SBLIs (Délery & Marvin 1986). The first DNS study of SBLI was reported by Adams (2000), who investigated the flow over a 18° ramp at free-stream Mach number $M_\infty = 3$, using Adams & Shariff's (1996) hybrid compact/ENO method. DNS of a 24° compression ramp configuration at $M_\infty = 2.9$ was performed by Wu & Martín (2007), using Taylor et al.'s (2007) bandwidth-optimized WENO algorithm. LES of the supersonic ramp flow was carried out by Rizzetta et al. (2001), using Visbal & Gaitonde's (2005) adaptive filtering technique, and by von Kaenel et al. (2004), using Adams & Stolz's (2002) regularization method.

Another frequently used prototype SBLI consists of the reflection of an oblique shock wave from a flat plate at which a boundary layer is developing. The first LES of impinging shock interaction was performed by Garnier et al. (2002), who used a baseline central fourth-order discretization augmented with a nonlinear WENO filter, with its local activation controlled by the Ducros sensor. Pirozzoli & Grasso (2006) performed a DNS study with flow conditions similar to Garnier et al.'s (2002), using a seventh-order WENO scheme. LES of the impinging shock

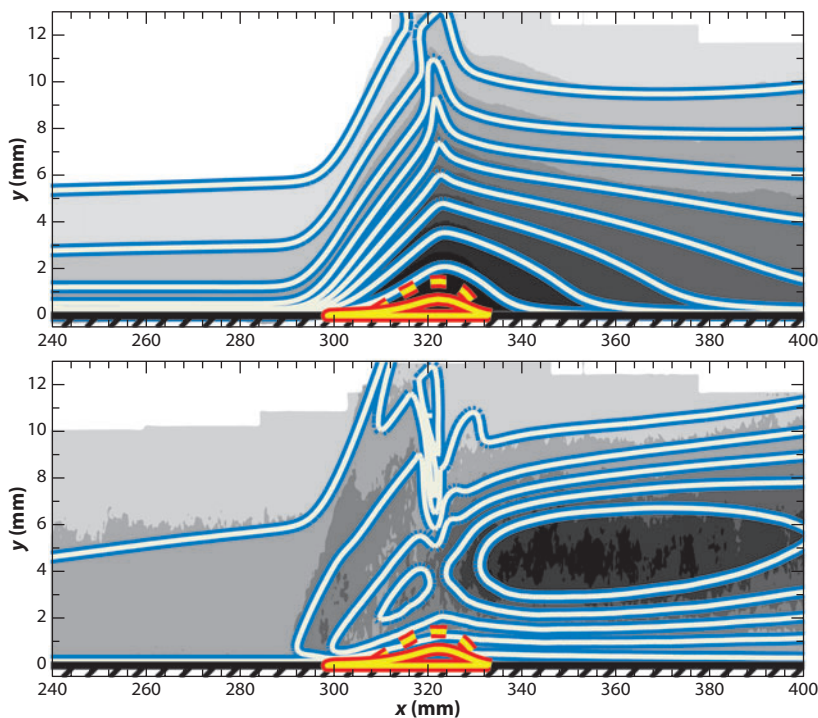


Figure 5

Large-eddy simulation (LES) of shock/boundary layer interactions (Touber & Sandham 2009) compared with particle image velocimetry (PIV) experimental data (Dupont et al. 2008). (*Top panel*) Mean streamwise velocity. (*Bottom panel*) Reynolds shear stress. The same contour levels are used for PIV (*filled contours*) and LES (*solid lines*). The yellow/red lines denote the mean separation line as from PIV (*dashed lines*) and LES (*solid lines*). Figure courtesy of E. Touber and N.D. Sandham.

interaction has also been performed by Toubert & Sandham (2009). Their numerical method relied on a baseline SHOEC scheme, and shock-capturing was achieved through a TVD-based filter controlled by the Ducros sensor. Sample results of Toubert & Sandham's (2009) calculations are shown in **Figure 5**, which compares the mean LES field (in terms of streamwise velocity and turbulent shear stress) and reference experimental data (Dupont et al. 2008). Overall, the LES results are in very good agreement with experimental PIV data, with the most apparent difference being the size of the separation bubble. However, the boundary layer thickening is well captured, as well as the amplification of the Reynolds shear stress past the interaction zone, which is associated with the shedding of vortices.

SBLIs also occur under transonic conditions. Sandham et al. (2003) performed the first LES of transonic SBLI over a circular-arc bump using the SHOEC scheme with nonlinear TVD filtering. Pirozzoli et al. (2010) have recently reported DNS results of transonic SBLI at $M_\infty = 1.3$ over a flat plate using a hybrid discretization of the convective fluxes. In **Figure 6** we report results obtained with an expanded version of the solver (M. Bernardini, S. Pirozzoli & F. Grasso, submitted manuscript), whereby smooth flow regions are handled by means of conservative sixth-order central discretization of the split convective form given in Equation 34, and shock waves are captured

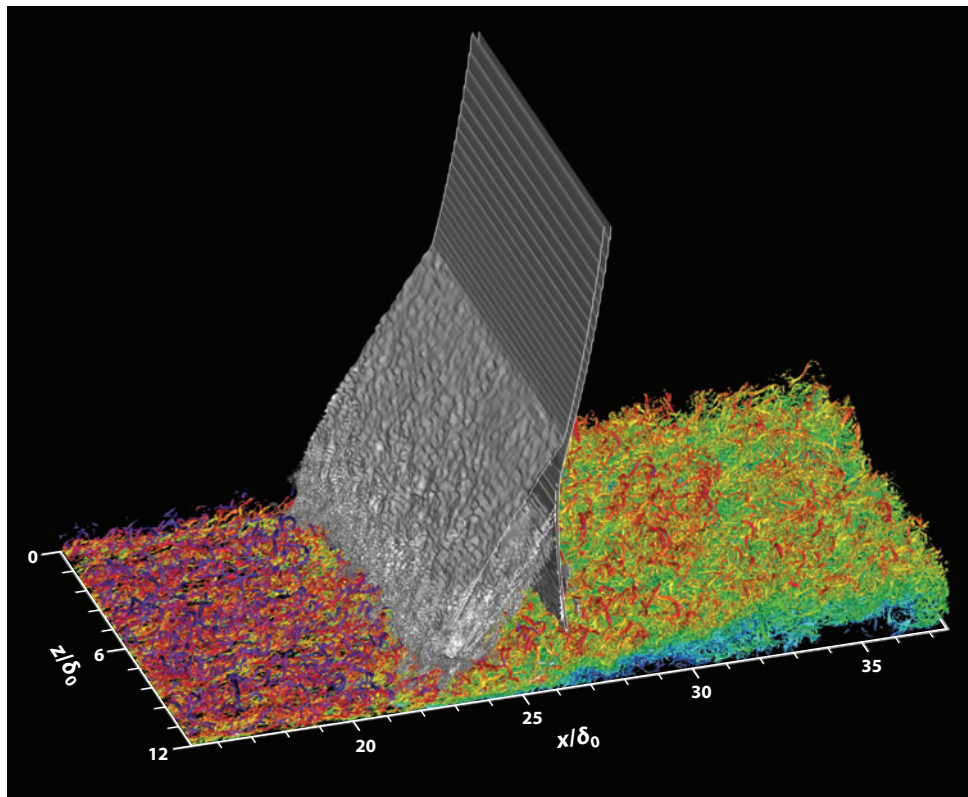


Figure 6

Direct numerical simulation of transonic shock/boundary layer interactions (M. Bernardini, S. Pirozzoli & F. Grasso, submitted manuscript). The shock system (*gray shading*) is identified as an isosurface of the modified Ducros sensor ($\Theta = 0.0025$). Vortical structures are deduced as isosurfaces of the swirling strength ($\lambda_{ci} = 1.6 u_\infty / \delta_0$) and are colored with the local Mach number (the levels are from 0 to 1.3, and the color scale from blue to violet). Axes are scaled with respect to the incoming boundary layer thickness (δ_0).

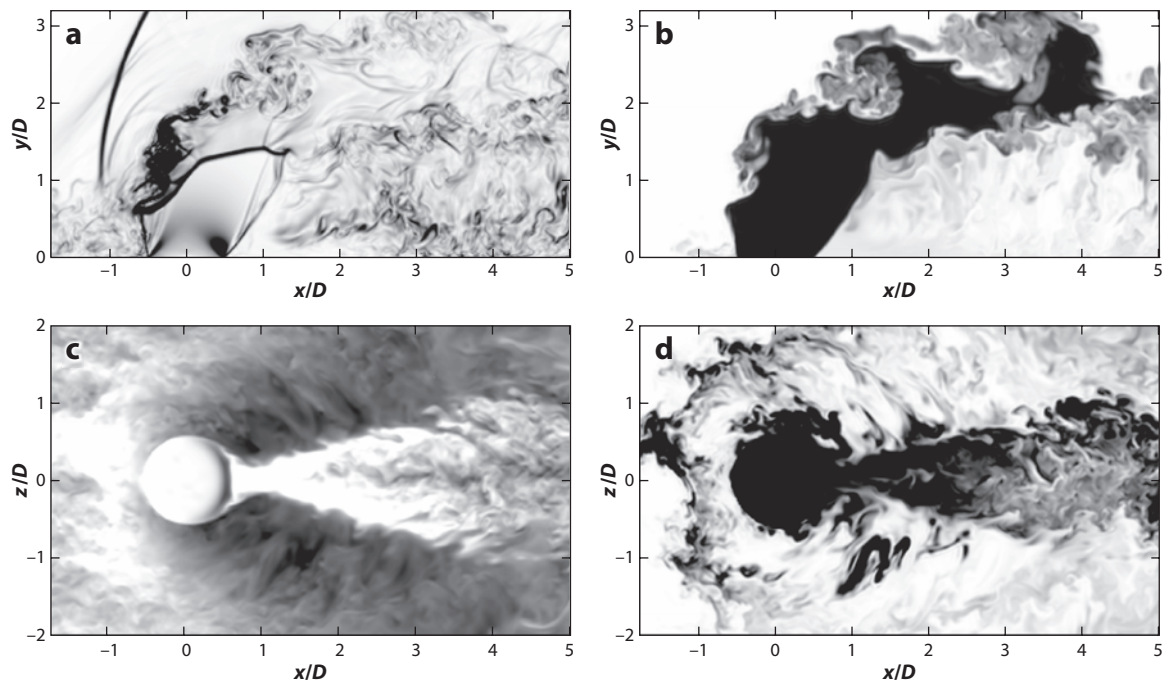


Figure 7

Large-eddy simulation of a jet in supersonic cross-stream (Kawai & Lele 2010). Visualizations in the streamwise/wall-normal plane are shown in the top row: (a) density gradient and (b) jet fluid. Visualizations in the wall-parallel plane are shown in the bottom row: (c) streamwise velocity and (d) jet fluid (*right*). D is the jet diameter. Figure courtesy of S. Kawai and S.K. Lele.

through a fifth-order WENO scheme, with the switch based on the modified Ducros sensor. In **Figure 6**, the shock system is visualized through the Ducros sensor and vortical structures through isosurfaces of the swirling strength (Zhou et al. 1999). The figure highlights the 3D nature of the lambda-like shock pattern, whereby the vortical structures in the incoming boundary layer cause the spanwise wrinkling of the upstream compression fan. Numerous hairpin-shaped vortex loops, resembling those found in incompressible boundary layer DNS, are observed both in the upstream boundary layer and past the interacting shock.

The turbulent mixing resulting from the injection of an underexpanded sonic jet in a supersonic cross-flow was studied by Kawai & Lele (2010) by means of LES. The authors used the artificial viscosity method in Kawai & Lele's (2007) version, employing sixth-order compact approximations of the spatial derivatives, coupled with eight-order low-pass filtering. The flow visualizations shown in **Figure 7** demonstrate the capability of the numerical method to capture the front bow shock, the upstream separation shock, the barrel shock, and the Mach stem, all without spurious wiggles, and at the same time to accurately resolve a broad range of turbulence scales.

Low-dissipative shock-capturing methods have also been used by Hill et al. (2006) to analyze Richtmyer-Meshkov instability with reshock. The authors used a DRP-like central approximation of the equations in convective split form in smooth regions and switched to the tuned-WENO scheme of Hill & Pullin (2004) near shock waves, with the switch controlled by the local curvature of the pressure and density field, in a fashion similar to the Jameson sensor.

Many other applications of low-dissipative shock-capturing algorithms are collected in the review paper by Ekaterinaris (2005).

6. CONCLUSIONS

Above we present some currently available numerical methods for high-speed flows, emphasizing FD schemes of current use for LES and DNS of compressible flows with shocks. Owing to their great computational efficiency, FD methods are preferable for large-scale computations whenever the physical problem allows the use of uniform Cartesian or smooth curvilinear meshes.

There are two classes of methods specifically designed for application to smooth and shocked flows. With regard to the former class, several mathematical principles can be exploited for the design of stable, low-dissipative schemes without reverting to upwinding or filtering. In particular, nonlinearly stable schemes can be designed by enforcing at the discrete level properties inherent to the Euler equations in the absence of shocks, namely preservation of the total kinetic energy and conservation properties related to the thermodynamic entropy. The results of several investigators, and numerical tests herein reported, support the effectiveness of these methods in providing nonlinear stability for the calculation of high-Reynolds number flows, without extra damping. A survey of the current literature leads the author to favor energy-preserving schemes that rely on the expansion of convective derivatives. Numerical schemes stemming from central explicit discretization of equations in convective split form are computationally efficient, have good nonlinear stability properties, and can be formulated in locally conservative form. In particular, the variant of convective splitting proposed by Kennedy & Gruber (2008) seems to be promising for smooth flows with strong density variations.

Regarding methods for shocked flows, WENO schemes (and their variants) seem to have superseded other shock-capturing methods in the past decade, having proven to be extremely accurate and robust in the presence of strong shock waves and complex shock interactions. WENO schemes, however, are quite CPU intensive and suffer from excessive numerical damping in smooth zones of the flow field. Therefore, their application to LES and DNS is suggested only in hybrid form, i.e., in conjunction with a nondissipative algorithm to treat smooth flow zones. Effective strategies for coupling shock-capturing and nondissipative methods include hybridization and nonlinear filtering. Both strategies rely on shock sensors that have to be as simple and effective as possible. One such choice is the Ducros sensor, which was found to perform reasonably well in many shock/turbulence interaction problems, particularly with some minor case-by-case modification and tuning of the threshold value for activation of shock-capturing. With regard to the hybrid methods, a suggested best practice is marking critical nodes using a shock sensor and then padding a sufficient number of nodes around the critical ones to make sure that the stencil of the underlying nondissipative scheme does not cross shocked zones.

An attractive alternative to hybrid shock-capturing methods is the nonlinear artificial viscosity methods, especially in their most recent variants (Mani et al. 2009). These methods have the advantage that they are applicable in the same form throughout the computational domain, without the need of hybridization, and that their implementation in existing compressible Navier-Stokes solvers for shock-free flows is straightforward. Applications of the method are recent, and further testing is required to demonstrate comparable reliability to WENO for a range of applications.

In all cases, a locally conservative discretization is encouraged. Although it is frequently stated that nonconservative methods are capable of handling shock waves properly, a locally conservative formulation (based on numerical fluxes) automatically guarantees global discrete conservation of integral properties through the telescopic property and the applicability of the Lax-Wendroff theorem. Note that there is no extra cost incurred with the use of a locally conservative discretization, and numerical fluxes are available for energy-consistent and entropy-consistent schemes (see Sections 2.3 and 2.4). Furthermore, hybridization with shock-capturing schemes (always formulated in conservation form) is more natural.

Some open issues remain. First, one must be aware that the global order of accuracy of shock-capturing schemes in unsteady problems is always reduced to unity, and shock-capturing is the cause of spurious oscillations, especially downstream of slowly moving shocks. These limitations, related to the misrepresentation of discontinuities on a mesh with finite spacing, can only be overcome by some form of shock-fitting. A detailed study of the effect of shock-capturing oscillations on the prediction of shock/sound and shock/turbulence interactions is lacking and would be highly desirable. Second, even though hybrid schemes are frequently used, a systematic quantitative analysis of the coupling between shock-capturing and nondissipative schemes has not been carried out [however, hints are found in the work of Larsson & Gustafsson (2008)]. Third, a comparative efficiency analysis of numerical algorithms (in terms of CPU cost for given error tolerance) for problems involving shock waves is not available at present, and cost figures are seldom reported in computational studies. Fourth, it appears that efficient, low-dissipative methods suitable for compressible turbulence simulation on unstructured meshes are lacking in the literature, one notable exception being the recent work of Subbareddy & Candler (2009) (however, limited to second-order accuracy). In this respect, alternative avenues to FV methods are also worth exploring (Dumbser et al. 2008). Further efforts are needed before computational gasdynamics can reach a fully mature stage and cope with the growing demand for DNS and LES of high-speed turbulent flows for configurations of technological relevance.

SUMMARY POINTS

1. Methods designed for smooth flows and shocked flows have quite different features. The former class is driven by the intent of achieving nonlinear stability without introducing numerical dissipation. The latter class attempts to stabilize shock computations through the addition of some (possibly not excessive) numerical dissipation.
2. Robust and accurate methods for smooth flows can be developed with the guidance of physical conservation principles of kinetic energy and entropy.
3. Discretization of the split convective form of the equations leads to methods that are nonlinearly stable for smooth flows also in the infinite-Reynolds number limit.
4. The expanded split convective form of Kennedy & Gruber (2008) shows potential advantages for flows with large density variations.
5. Shock-capturing methods are always globally first-order accurate for unsteady problems, as the shock location is unknown to $O(b)$, where b is the mesh spacing.
6. WENO schemes are the currently dominant class of shock-capturing methods, as they are at the same time accurate and robust.
7. WENO schemes are too dissipative for DNS and LES and should be preferably used in hybrid form, i.e., in conjunction with low-dissipative algorithms for smooth parts of the flow.
8. A key role for the success of hybrid methods is played by shock sensors that should be able to localize the necessary amount of numerical dissipation around shock waves.
9. Nonlinear artificial viscosity methods constitute an attractive alternative to hybrid-WENO schemes.

FUTURE ISSUES

1. Shock-capturing schemes generate spurious oscillations downstream of slowly moving shocks, whose influence on the prediction of shock/sound and shock/turbulence problems is not known.
2. The performance of shock-capturing schemes for smooth flow cases has not been completely characterized.
3. An objective comparison of the efficiency of shock-capturing schemes would be desirable.
4. The errors associated with the hybridization of low-dissipative and shock-capturing schemes should be more thoroughly investigated.
5. Researchers should seek out efficient high-order, low-dissipative methods for DNS and LES in complex geometries.

DISCLOSURE STATEMENT

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